Math 252
Spring 2023

Name ______________________

Test 1

1

a) Show how to calculate $\binom{128}{3} = C(128, 3)$ **without a calculator.** You don't have to do the arithmetic, just show the arithmetic that needs to be done. But it has to be arithmetic that you could do without a calculator in just a couple minutes.

b) Write a counting problem for which $\binom{128}{3}$ gives the answer. (Keep your problem simple, but be creative: **do not use donuts, bitstrings, or cookies**.)

2 Draw a graph with 6 vertices, each of which has degree 3.

3 Explain why it is impossible to draw a graph with 7 vertices, all of which are degree 3.

4 Below is an adjacency matrix representation of a graph G.

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

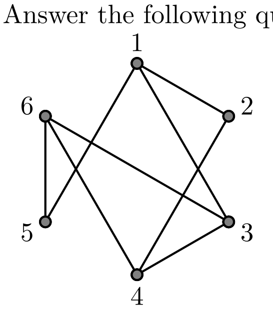
a) Is G directed or undirected? How can you tell?

b) How many vertices does G have? How can you tell?

c) How many edges does G have? How can you tell?

d) Does G have any self-loops? How can you tell?

5 Answer the following questions about the graph below.



a) What is an Euler circuit? Does this graph have an Euler circuit? How do you know?

b) Does the graph have an Euler path that is not a circuit? How do you know?

c) What is a bipartite graph? Is this graph bipartite? How do you know?

d) What is a planar graph? Is this graph planar? How do you know?

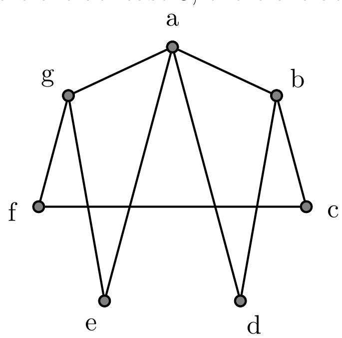
6 For **one** of K_5 or $K_{3,3}$ (your choice),

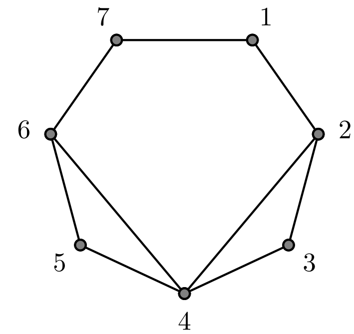
a) Draw the graph.

b) Explain how we know that the graph cannot be drawn in the plane with no edge crossings. **Do not use Kuratowski's Theorem or any result we did not cover in class.**

7 **How many isomorphisms** are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There are at least 3; there are at most 2, etc.)





8 A security system requires its users to select a 5-digit PIN code. This is a code consisting of a sequence of 5 digits (0-9). Examples: 01235, 82071, 13317, 70427.

a) How many such PIN codes are there?

b) How many PIN codes do not repeat any digits? (Examples: 01235, 82071.)

c) How many PIN codes have *exactly* 3 even digits (0, 2, 4, 6, 8)? (Examples: 82071, 70427.)

d) How many PIN codes have *exactly* 3 even digits (0, 2, 4, 6, 8) and also do not repeat any digits? (Example: 82071.)